Time, Openness and Invariance

The K_{frak} recursion enforces a boundaryless, open time structure — unrolled, continuous, curved and scale-invariant

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Observer, scale and the simplicity of the core

Every observer sits on a scale. They do not see the object — they see the *projection* of the object onto their scale. Projections from different scales must differ, not because the observers err, but because the generative object is richer than any single view. The perspective distortion is not a mistake — it is structural, the unavoidable consequence of the object (T^4, compact, boundaryless) not fitting into any single projection. Hardly any scale can be surveyed simultaneously with another; the results of observation therefore already differ perspectively as soon as a further observer is added.

This is the deeper reason why quantum mechanics and the theory of relativity are so difficult to reconcile: they are projections onto different scales of the same object, and of course they look different — not because one is wrong, but because neither is complete.

All the more remarkable, therefore, is what follows: that the core is so simple. One parameter $\xi = 4/30000$. One relation $\tilde{T} \cdot m = 1$. One compact torus $T^4/\mathbb{Z}_3$. From these fall the fine-structure constant α, the lepton masses, the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ — the entire complexity of the Standard Model, the whole richness of the phenomena that appear so overwhelming on the various observer scales. The complexity does not reside in the core — it resides in the projections. The core is simple.

This is not self-evident. It could equally have been that nature starts afresh on each scale, that there is no common root, that the variety is irreducible — that would be the generic outcome. That a single structure instead carries all scales becomes visible only when one stops treating one's own scale as the only possible one.

Reduction is the condition of evaluability

A projection is always a reduction — it leaves out. That is the loss. But precisely by leaving out, it makes a partial domain *evaluable* that would have been submerged in the full object. The full T^4 object carries all symmetries simultaneously; that is exactly why it is not directly evaluable as a whole. Only the reduction to a scale isolates a symmetry far enough that one can compute with it, measure it, derive something from it. Reduction and focus are the same thing.

This is not a makeshift but the mechanism of cognition: every evaluation needs a focus, and every focus is a reduction. The particle physicist who measures α evaluates the finest scale because they have projected away everything coarser; the cosmologist who measures the redshift evaluates the coarsest because they have integrated away everything finer. Both gain their evaluation through what they leave out.

The decisive point: the new evaluations are not already contained in the full object — they arise at the projection level. A phonon does not exist at the particle scale; it is an evaluation that becomes possible only on the crystalline projection. An orbital is not a property of T^4 but an evaluation at the atomic scale. The projection *generates* evaluable structure that would not exist without the reduction. This is why each new scale brings forth new physics: not because new objects are there, but because new evaluations become possible. A projection that left out nothing would evaluate nothing.

FFGFT's own method is a special case of this: $\theta = 2/9$ becomes evaluable by projecting onto the $\mathbb{C}^3$ fibre (Hilbert-space transformation, Doc. 230/282). On T^4 directly it was hidden; only the reduction to the fibre made it visible as the diagonalisation output of the $\mathbb{Z}_3$ circulant (Doc. 291). The reduction was not a loss but the condition under which the structure became legible at all.

From body to society — the same structure at every level

That observers on different scales see different projections is not merely a physical abstraction — it begins with the simplest biological perception. Two eyes are already two observers at slightly different positions. The separation is small but real; the brain *constructs* a three-dimensional world from two slightly different projections, a world that exists nowhere directly in that form. Depth is a reconstruction, not direct perception. Further sense organs — proprioception, balance, touch, hearing — each contribute their own scale and their own projection, and the synthesis of all of them feels like immediate reality, even though it is a highly processed achievement.

This has a direct consequence for physics: that we *experience* the world as three-dimensional is not evidence that it fundamentally is. It is the finding that three dimensions are the projection onto which our sensory apparatus has been optimised. T^4 beneath this does not contradict the three-dimensional experience — it is the reason why the projection is so stable and consistent.

The same structure continues one level higher. A society is a collection of observers, each on their own scale, each with their own projection. What arises as social reality — norms, values, worldviews, conflicts — is the synthesis of millions of slightly different projections, none of them complete, none of them wrong in any absolute sense, but each perspectively distorted. And as with binocular vision: the synthesis feels like direct perception, even though it is a construction.

This explains several things that would otherwise be puzzling. Conflicts are so persistent because two observers on different scales see the same object and reach different conclusions — both correct for their own projection, both incomplete. As long as no common higher-level scale is accessible, the conflict cannot be resolved by argument alone; one would have to shift the projection plane, and that is the most difficult thing there is. Historical change is therefore so slow: a society has calibrated its synthetic achievement to a particular scale, which works well within that scale and generates blindness to others. Paradigm shifts are scale shifts, and scale shifts are painful because the previous coherent picture dissolves before the new one has formed.

Science is the systematic attempt to triangulate projections — multiple observers, multiple scales, contradictions treated as signal rather than noise. This is not self-evident; it runs against the natural impulse to take one's own projection for reality.

The arc back to the core: that despite the multiplicity of projections — biological, physical, social — something invariant underlies them all, something that shows itself completely in no single projection but shines through in all of them, is not self-evident. The generic outcome would be that each scale had its own irreducible truth. That there is instead a common core — in physics ξ on T^4, in experience the synthetic achievement that carries all projections — is the actual surprise, at every level.

Cognition: recursive layering and sensory binding

The biological level leads to a foundation stone already laid in the discussion of consciousness: no consciousness without recursion and without sensory input. Both conditions are necessary, and both are FFGFT structural elements.

Without recursion there is no consciousness: it requires the recursive layering that applies to itself — self-reference, accumulating memory, a structure that carries its own history. This is the same generative form as the K_{frak} recursion from which the scale sequence arises: one level generates the next by the same rule. A mere sliding memory window that discards older states does not suffice — the actual achievement is accumulation, the structural carrying of history, not its replacement.

Without sensory input there is no consciousness: cognitive processes never operate unbound. They are always coupled to a substrate that takes in projections of the world through sensory channels. This is the observer binding of this document: not a free-floating process but an observer bound to a scale, receiving the world as a projection.

Thus the two conditions without which there is no consciousness — recursive layering and sensory binding — are exactly the two structural elements FFGFT carries anyway: the recursion and the bound projection. This is *not* a derivation of consciousness; no such claim is made here and it would lie beyond what is shown. It is a convergence at the level of *necessary conditions*: a consciousness-capable system must be recursively layered and substrate-bound, and both are native to FFGFT.

Two delimitations keep this honest. *First*, cognitive and physical processes share the generative form (recursive layering), but on the cognitive side there is as yet no measurable, quantitative recursion law — no "ξ of cognition", no pitch, no ratio e. The form is shared, not (yet) the law. *Second*, the absence of such a law is no evidence against a common nature of the processes: missing measurability speaks not against the underlying structure but only against its present accessibility — as an inadequate measurement technique never measures against what it cannot yet resolve. The question therefore remains open upward: plausibly suggested by the FFGFT structure (everything as a projection of the same recursive object), not refuted, only not shown. The foundation stone — a shared necessary form — stands; the full connection is programme.

Scale sequence as a succession of projection shifts

From particle physics through atomic, crystalline and biological to galactic structures, the same generative object appears fundamentally different at each level — not because the objects differ, but because each scale makes a different symmetry of the core visible while integrating the others away.

At the *particle scale* the projection cuts most finely: the $\mathbb{Z}_3$ symmetry, the SU(3) colour structure and the discrete masses are direct imprints of the T^4 geometry. The fine-structure constant α and the phase $\theta = 2/9$ are most directly accessible here — the core is nearest on this scale. At the *atomic scale* ($\sim a_0$) the colour structure has already been integrated away and appears as neutral; what becomes visible are electromagnetic resonances — orbitals as standing waves, discrete energy levels as torus geometry on this coarser scale. At the *crystalline scale* ($\sim$ Å–μm) the translation symmetry of the lattice comes to the fore; collective modes (phonons, band structures) arise as new effective degrees of freedom that do not exist at the particle scale. At the *biological scale* (nm–m) broken symmetries dominate — chirality, self-replication, information processing; life is a coherence window, a scale at which a particular kind of complexity can just remain stable, because below it thermal fluctuations and above it gravity force other projections. At the *galactic scale* (kpc–Mpc) everything except gravity has been integrated away; the spiral structure of galaxies carries the same logarithmic self-similarity as the $\mathrm{K}_{\mathrm{frak}}$ recursion — not coincidentally, but as an expression of the scale invariance of the core at this projection level.

Scale transitions and their anomalies

The transitions between scales are particularly instructive — precisely because anomalies appear there that cannot be explained from within a single scale. A transition is where the previously dominant symmetry tips and a new one comes to the fore; at this tipping point the system is simultaneously shaped by both projections and fully described by neither. This structurally generates anomalies: deviations that appear as disturbances from one side and as harbingers of a new order from the other.

Familiar examples include phase transitions (critical fluctuations whose correlation lengths diverge — the system "senses" on all scales simultaneously that a transition is imminent), the transition from classical to quantum mechanics (decoherence as projection stabilisation, not as fundamental collapse), or the transition from biological to social organisation (emergence of collective intelligence not foreshadowed at the single-cell scale). In all cases the anomaly lies *in the transition*, not within either scale itself — and is therefore not fully explicable from any single scale.

For FFGFT this is a directed pointer: the places where a scale's standard description reaches its limits — unresolved anomalies, unexplained constants, emergent phenomena without derivation — are precisely the places where a projection shift occurs. They are not errors of the theory but signatures of the transition.

Speculation: unobservable intermediate ranges

The scales considered so far are those accessible to us directly or indirectly — through instruments, through consequences, through indirect inference. But the fact that scales are observer perspectives admits a further thought, which is explicitly booked here as *speculation* (P35: not a derived finding but an open question):

Between the observable scales there may lie ranges that are in principle inaccessible to us — not for technical reasons but structurally, because none of our sense organs or instruments projects onto these intermediate scales. Projections are selective; they make one symmetry visible and suppress others. It is not excluded that there are scales on which coherent structures exist that leave no direct imprint in any of our accessible projections — at most an indirect one, as an anomaly at the edges of neighbouring scales.

This would not be mystical but structurally consistent: if the generative object T^4 is richer than any projection, and if projections are always selective, then the generic outcome is that there are regions of the object captured by none of our projections. Dark matter and dark energy would, in this reading, not be new substances but signatures of such intermediate ranges — structures at scales onto which our instruments do not project, but which make themselves felt as anomalies at the edges of the galactic and cosmic projections.

This remains speculation. But it is directed speculation: it names precisely where to look for evidence — at the transitions, in the anomalies, at the places where a scale reaches its explanatory limits without internal reason.

Starting point: a recursion with no reachable boundary

The K_{frak} recursion

$$\xi_{n+1} = \xi_n\,(1 - 100\,\xi_n) \tag{1}$$

does not run endlessly in the sense of uniform repetition — it asymptotes. ξ_n decays monotonically towards zero, but never reaches zero in finitely many steps. This is not a gap in the construction; it is its statement: the sequence is infinitely long, but the gaps between successive steps shrink exponentially, precisely by the factor $1/e$ per 2π turn.

What does this mean for experienced, measurable time? The measurable time step is not n but $\rho = 1/(100\,\xi_n)$ — and this reciprocal *grows*. The further the recursion proceeds, the larger ρ becomes and the coarser the time scale on which something changes. Experienced from within, one does not encounter a uniform sequence but a *slowing* dynamics: early steps are dense and fast (small ρ, large ξ, sub-Planck-proximate), late steps are wide and sluggish (large ρ, small ξ, macroscopic). The experienced time is the integral over ρ — and this integral grows logarithmically: $D(k) = \sum 100\,\xi_n \to \ln(1 + k/75)$. The cumulative defect *is* the measurable time evolution (Doc. 295).

From this it follows immediately: experienced time has no end. ρ grows without bound, $D(k)$ diverges, ξ_n never reaches zero. There is no final step, no endpoint, no "afterwards". And from the other side — the inner side — the same holds in mirror image: ξ never reaches its starting value ξ_0 from below, ρ never reaches $75 = 1/(100\,\xi_0)$ from above. No beginning, no end — both sides are asymptotic limits, not reachable boundaries.

Unrolled — temporally continuous — space curved — invariant

These four properties are not accidentally connected; they follow from one another.

Unrolled. The spiral unrolls the scale structure of T^4 into a directed time coordinate. Not because time "really is" a spiral, but because the projection — the observer's position on the spiral — singles out a direction. The unrolling is the transition from the compact picture to the open one: the same object, read from a particular scale.

Temporally continuous. This follows necessarily. The Hilbert-space bijection $H = L^2(T^4) \otimes \mathbb{C}^3$ is lossless (Doc. 230/282, bijection type III, Doc. 270); what is discretely open remains continuously open — otherwise the translation would not be faithful but distorting. The memory kernel (Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega - \omega_k)$ of the T^4 connection-Laplacian, Doc. 283) is defined on $\mathbb{R}$, not on an interval. No boundary on the left, no boundary on the right. Revivals without end, BLP backflow without closure — the same open structure, now in the continuous picture.

Space curved. This is the T^4 core itself. The curvature is not inserted after the fact; it is the generative structure: ξ on $T^4/\mathbb{Z}_3$ *is* the curvature from which the lepton masses, α, and $\theta = 2/9$ follow (Doc. 006/011/291/293). The geometry is not background — it is the object. Compact means finite volume but *no boundaries*: a geodesic on the torus runs without end, never encountering an edge.

Invariant. This is the deepest of the four points. The logarithmic spiral does not change its size (Doc. 295): it is self-similar — rotated by one turn and scaled by e, it lies congruent with itself. Rotation and scaling are *the same* operation for it. "Inward by $1/e$" and "outward by e" do not describe something growing or shrinking; they only indicate which scale one is currently examining. The object is invariant under this motion — there is no absolute size, only ratios, and ratios are the intrinsic quantities under $\tilde{T} \cdot m = 1$. What changes is the *observer scale*, not the object.

No beginning, no end — and no coincidence problem

The four properties together yield a consistent picture: an invariant, curved, boundaryless object — unrolled into an open, continuous time coordinate through the act of projection. No beginning, no end, no absolute size, no inserted curvature.

This differs structurally from models that require boundaries. A beginning at $t = 0$ is not a prediction but a limit at which the description ceases to hold — the singularity is not a result but an admission. A finite time interval requires a starting point, and the starting point demands a justification: why now, why this state, why these initial conditions. The coincidence problem, the flatness problem, the fine-tuning of initial conditions — all arise because a finite interval has a distinguished location.

FFGFT has no distinguished location. Every observer sits somewhere on an open trajectory; "now" is the current scale, not a privileged moment. The experienced time is not an interval between two points — it is an open trajectory on a boundaryless object.

New problems as a sign of progress

Such a view generates new questions — this is to be expected and is not an objection. An approach that raises no new problems usually also explains nothing new; it is without consequence. The decisive distinction is what *kind* of problems arise: whether they are *directed follow-on questions* (what still needs to be computed?) or genuine breaks (does $\tilde{T} \cdot m = 1$ break?).

The exchange is asymmetric in FFGFT's favour: in place of unexplained stipulations — why these masses, why $\alpha = 1/137$, why $\theta = 2/9$, where does non-locality come from — there appear consequences of a derivation, consequences that can be posed and whose resolution advances the programme. New problems arising from a deep solution are the signature of genuine progress, not of its absence. The task is not to avoid them but to record clearly which are directed follow-on questions — and so far, all of them are.

Note (P35): The statements in this document are structural in nature — they follow from $\tilde{T} \cdot m = 1$, the recursion, and the compactness of T^4. The cosmological transfer (open time, no singularity) is a *consequence* of this structure, not a standalone cosmological model. As a consequence it must be free of contradiction with the data — not complete in the sense of covering all phenomena of a dedicated model. This distinction between freedom from contradiction and completeness is to be maintained.